Introduction to Statistics in Data Science

Statistic is the branch of mathematics that deals with collecting, organizing, analyzing, interpreting, and presenting data.

What is Statistics ?

Main Objectives of Statistics

1.Describe data (using charts, averages, etc.)

2.Make decisions or predictions based on data

3.Identify patterns and relationships

4.Test hypothesis or assumptions

Types Of Statistics:

• Descriptive Statistics: Descriptive Statistics focuses on summarizing and presenting data. This involves measures like mean, median and standard deviation. It helps to understand the basic features of a dataset.

Example: If you have the ages of 5 students: 20, 21, 22, 23, 24, then:

•Mean age = (20+21+22+23+24)/5 = 22

• Inferential Statistics: Inferential Statistics involves drawing conclusions from sample data. This includes hypothesis testing and confidence intervals. It allows for generalizations about a larger population.

Example: You survey 100 people about internet usage and predict trends for the whole country.

Key Statistical Concepts:

1. Probability Distributions: Understanding the likelihood of different outcomes.

2. Central Tendency: Measuring the center of a dataset(mean, median and mode).

3. Measures of Dispersion: Assessing the spread of data(variance, standard deviation).

4. Hypothesis Testing: Evaluating claims about populations based on sample data.

Data Collection and Sampling :

• Surveys gather data through questionnaires and interviews.

• Experiments involve controlled manipulations to observe effects.

• Observational studies collect data without intervention.

Random sampling is used to avoid bias. Stratified and cluster techniques are common. Representative samples are crucial for valid conclusions.

Exploratory Data Analysis(EDA):

1)Visualizing Data

Use histograms, scatter plots and box plots to understand data.

2)Identifying Patterns

Discover trends and outliers in the data

3)Statistical Software

Utilize tools like Python, R and Excel for analysis.

Sample Data Statistical Inference :

Make predictions from sample data using confidence intervals and p- values. A/B testing helps in business decisions. Predict customer churn using relevant data.

Use Of Statistics in Data Science :

Statistics is the foundation of data science. It helps in understanding data, drawing conclusions, and making informed decisions. Here's where statistics is used in data science, with examples:

1.Data Collection & Sampling

Use: To design surveys or experiments and select representative data samples.

Example: Using random sampling to select a subset of customers for feedback instead of surveying all.

2. Data Summarization (Descriptive Statistics)

Use: To describe the main features of data.

Tools: Mean, median, mode, variance, standard deviation, range, percentiles.

Example: Finding the average income of customers or the standard deviation in sales.

3. Data Visualization

Use: Helps visually understand data distributions and relationships.

Tools: Histograms, boxplots, scatter plots.

Example: A boxplot showing customer spending across age groups.

4. Probability

Use: Models uncertainty and randomness in data.

Example: Estimating the chance that a customer will buy a product after visiting a website.

5. Statistical Inference

Use: Drawing conclusions about a population from a sample.

Tools: Hypothesis testing, confidence intervals.

Example: Testing if a new marketing strategy increases sales using a t-test.

6. Correlation & Regression

Use: Understanding relationships between variables.

Tools: Pearson correlation, linear regression, logistic regression.

Example: Predicting house prices based on size, location using linear regression.

7. Experimental Design (A/B Testing)

Use: Compare two versions of a product/feature to see which performs better.

Example: Testing two website layouts to see which one gets more clicks.

8. Feature Selection & Engineering

Use: Use statistical measures to choose relevant variables for models.

Example: Removing features with low variance or using correlation to avoid multicollinearity.

9. Model Evaluation

Use: Statistical metrics assess model performance.

Tools: Accuracy, precision, recall, F1-score, ROC curve, confusion matrix.

Example: Evaluating a spam detection model using precision and recall.

10. Time Series Analysis

Use: Analyze and forecast data over time.

Tools: Moving average, ARIMA, seasonality.

Example: Forecasting sales for the next month using historical data

Measure Of Central Tendency :

These describe the center point or typical value of a dataset.

a. Mean (Arithmetic Average) - Sum of all values divided by the number of values.

Mean = (x1 + x2 + ... + xn) / n

b. Median - The middle value when data is sorted.

- Odd Number of Values: If the dataset has an odd number of values, the median is simply the middle value
- Even Number of Values: If the dataset has an even number of values, the median is calculated as the average of the two middle values.

c. Mode - The value that occurs most often.

Use Cases of Mean, Median and Mode in Data Science:

- Mean = Useful when data is symmetrically distributed (no extreme outliers).
- Median = Best when data is skewed or contains outliers.
- Mode = Useful for categorical data or finding the most common value.

Measure of Variability:

These tell us how spread out the data is.

a. Range -- Maximum value - Minimum value

Range = max(data) - min(data)

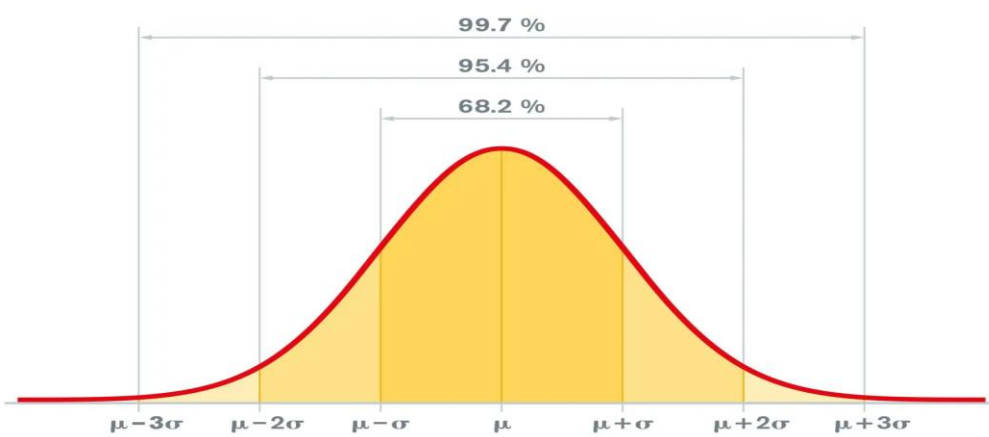
b. Variance -- Variance is the average of the squared differences between each data point and the mean of the dataset.

Variance = sum((xi - mean)^2) / n

c. Standard Deviation --- Square root of variance. It's a measure of the average distance of the data points from the mean.

Variance and Standard Deviation Formula

	Population	Sample
Variance	$\sigma^2 = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$	$S^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}$
Standard Deviation	$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$	$S = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}}$



Measure of shape :

1. Skewness - Skewness measures the asymmetry of a data distribution around the mean.

Types:

- Skewness = 0 → Symmetrical (like normal distribution)
- Skewness > 0 → Right-skewed (tail on the right)
- Skewness < 0 → Left-skewed (tail on the left)

Skewness indicates data imbalance

2. Kurtosis - Kurtosis measures the tailedness or peakedness of a distribution.

Types (excess kurtosis):

- Kurtosis ≈ 0 → Mesokurtic (normal curve)
- Kurtosis > 0 → Leptokurtic (heavy tails, sharp peak)
- Kurtosis < 0 → Platykurtic (light tails, flat peak)

Leptokurtic: Data with more outliers (e.g., financial returns)

Platykurtic: Data with fewer outliers

Helps detect outliers.